A MORPHISM OF ALGEBRAIC SPACES IS UNRAMIFIED EXACTLY WHEN ITS DIAGONAL IS AN OPEN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Both directions of the equivalence are written out by the author, with the cartesian diagram as the working object; the local finite type hypothesis in the second direction is part of the statement span.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The diagonal of a morphism of algebraic spaces is a representable monomorphism. Lemma lemma-properties-diagonal, used at proof lines 7737-7738. Supplies the input that has to be upgraded to an open immersion.
- Etaleness is local on the source and is preserved by etale base change of schemes over Y. Properties of Spaces, Lemmas spaces-properties-lemma-etale-local and spaces-properties-lemma-etale-permanence, used at proof lines 7752-7767 and 7780-7784. Transfers the property of the diagonal between the scheme and the algebraic space picture.
- The scheme case: unramified equals open diagonal. Morphisms, Lemma morphisms-lemma-diagonal-unramified-morphism, used at proof lines 7761-7763 and 7789-7790. Is the input applied to U -> V in both directions.
- A representable etale monomorphism is an open immersion. Etale Morphisms, Theorem etale-theorem-etale-radicial-open, with Spaces, Lemma spaces-lemma-representable-transformations-property-implication, used at proof lines 7768-7774. Converts the etale monomorphism into an open immersion.
- The diagonal of a morphism of schemes is an immersion. Schemes, Lemma schemes-lemma-diagonal-immersion, used at proof lines 7785-7787. Provides the immersion half in the converse direction.

Source reference. Original theorem 025G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2813–2813. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4570–4570. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3225–3225. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01KJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2052–2052. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3883–3883. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03FS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4503–4503. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03FV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4506–4506. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S.*

(1) *If f is unramified, then the diagonal morphism $\Delta_{X/Y} : X \to X \times_Y X$ is an open immersion.*
(2) *If f is locally of finite type and $\Delta_{X/Y}$ is an open immersion, then f is unramified.*

Proof. We know in any case that $\Delta_{X/Y}$ is a representable monomorphism, see Lemma 03HK. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme U and a surjective étale morphism $U \to X \times_Y V$. Consider the commutative diagram

$$\begin{array}{ccccc} U & \xrightarrow{\Delta_{U/V}} & U \times_V U & \longrightarrow & V \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle \Delta_{V/Y}} \\ X & \xrightarrow{\Delta_{X/Y}} & X \times_Y X & \longrightarrow & V \times_Y V \end{array}$$

with cartesian right square. The left vertical arrow is surjective étale. The right vertical arrow is étale as a morphism between schemes étale over Y, see Properties of Spaces, Lemma 03FV. Hence the middle vertical arrow is étale too (but it need not be surjective).

Assume f is unramified. Then $U \to V$ is unramified, hence $\Delta_{U/V}$ is an open immersion by Morphisms, Lemma 02GE. Looking at the left square of the diagram above we conclude that $\Delta_{X/Y}$ is an étale morphism, see Properties of Spaces, Lemma 03FS. Hence $\Delta_{X/Y}$ is a representable étale monomorphism, which implies that it is an open immersion by Étale Morphisms, Theorem 025G. (See also Spaces, Lemma 02YO for the translation from schemes language into the language of functors.)

Assume that f is locally of finite type and that $\Delta_{X/Y}$ is an open immersion. This implies that $U \to V$ is locally of finite type too (by definition of a morphism of algebraic spaces which is locally of finite type). Looking at the displayed diagram above we conclude that $\Delta_{U/V}$ is étale as a morphism between schemes étale over $X \times_Y X$, see Properties of Spaces, Lemma 03FV. But since $\Delta_{U/V}$ is the diagonal of a morphism between schemes we see that it is in any case an immersion, see Schemes, Lemma 01KJ. Hence it is an open immersion, and we conclude that $U \to V$ is unramified by Morphisms, Lemma 02GE. This in turn means that f is unramified by definition. □